they developed at the Pressure Transducer Unit (PTU) in Bangalore. PTU, where we made transducers for the French with their technology at lower cost, was a spinoff of the Vikas-Viking barter deal, something which proved to be a bonus for us.

For the engine to be part of the first PSLV flight, we continued to source parts from the original suppliers of SEP, Vernon. A French company called Le Joint de Francis supplied 'O' rings; Simmonds made fasteners. 'Make in India' became an official slogan of the government only in September 2014 after Narendra Modi became the Indian Prime Minister, but our Vikas indigenisation drive saw some of the foreign manufacturers making components for our rocket engine as early as in the 1980s, in India. One such was SNFE India Pvt Ltd, the Indian arm of a French company that made bearings.

Separate teams worked on fill-and-drain systems, pressurisation of tanks, gimbal actuator, roll control system and 'pogo and sloshing' control (to prevent the fuel and oxidiser from sloshing in the tank to create instability). Precision was the catchphrase and motto, for a minor mass difference or structural deviation can send the rocket haywire. One of the crucial aspects here was balancing the turbo pump assembly. BHEL, Bhopal did a good job with this.

We were also working on mock-ups, the making of models and assembling them to get a hang of handling components. It gave us a feel of what kind of space is needed, how the cranes need to be manoeuvred to move a component and stack it on another on the work floor.

This is when my colleague P S Krishnamoorthy brought P Ravindra Reddy, an engineer-entrepreneur from Hyderabad to meet me. Reddy was the founder chairman of MTAR Technologies, which was already doing some work for the Department of Atomic Energy.

Krishnamoorthy told me that Vikram Sarabhai had a good opinion of Reddy who started off with a small workshop in Hyderabad's Bala Nagar industrial area and expanded his factory systematically, with a passion for indigenisation. What is more, he was an engineer. That was an impressive introduction, but I had to convince myself of Reddy's abilities before I gave